



Trends in employment data

References:

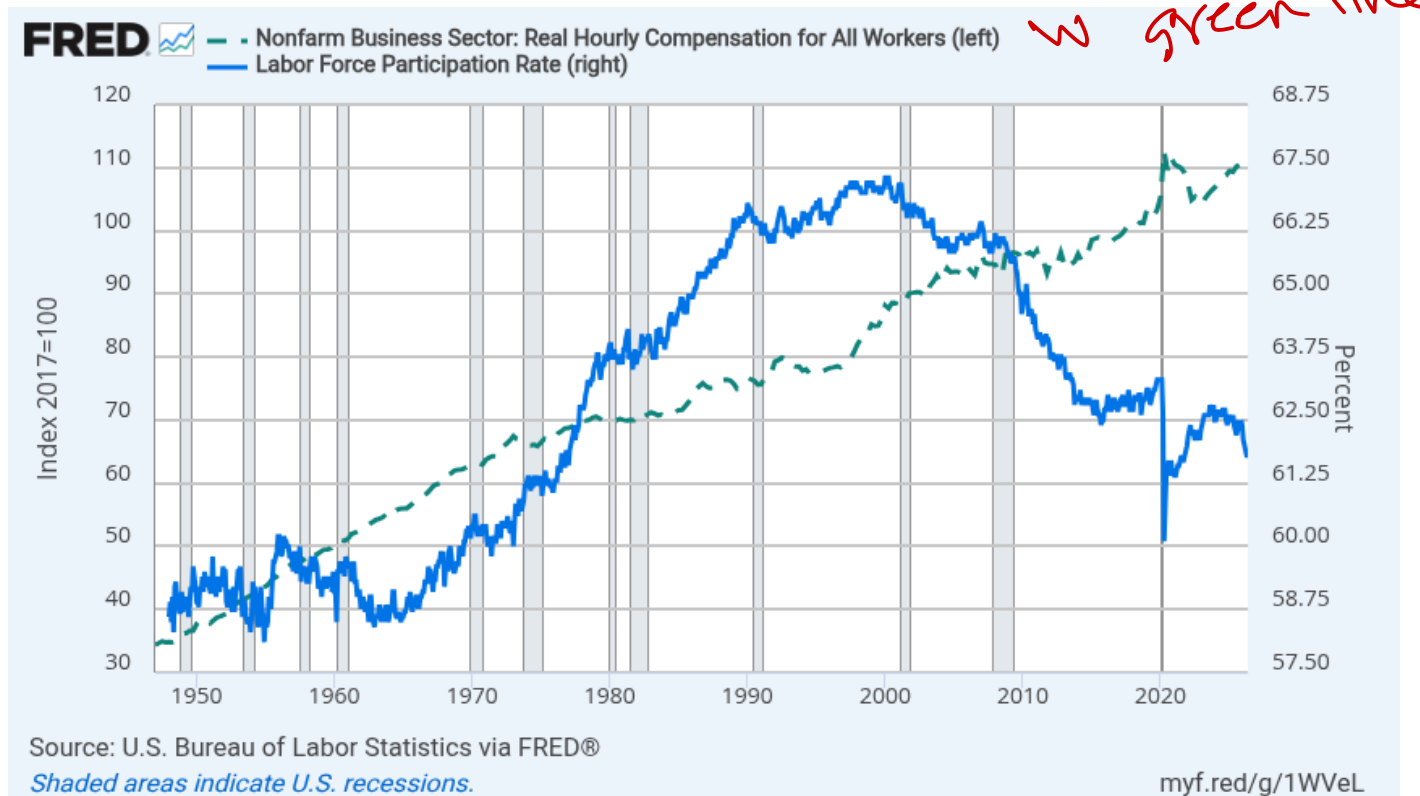
Chapter 4, Williamson

McGratten and Rogerson, *Changes in Hours Worked*

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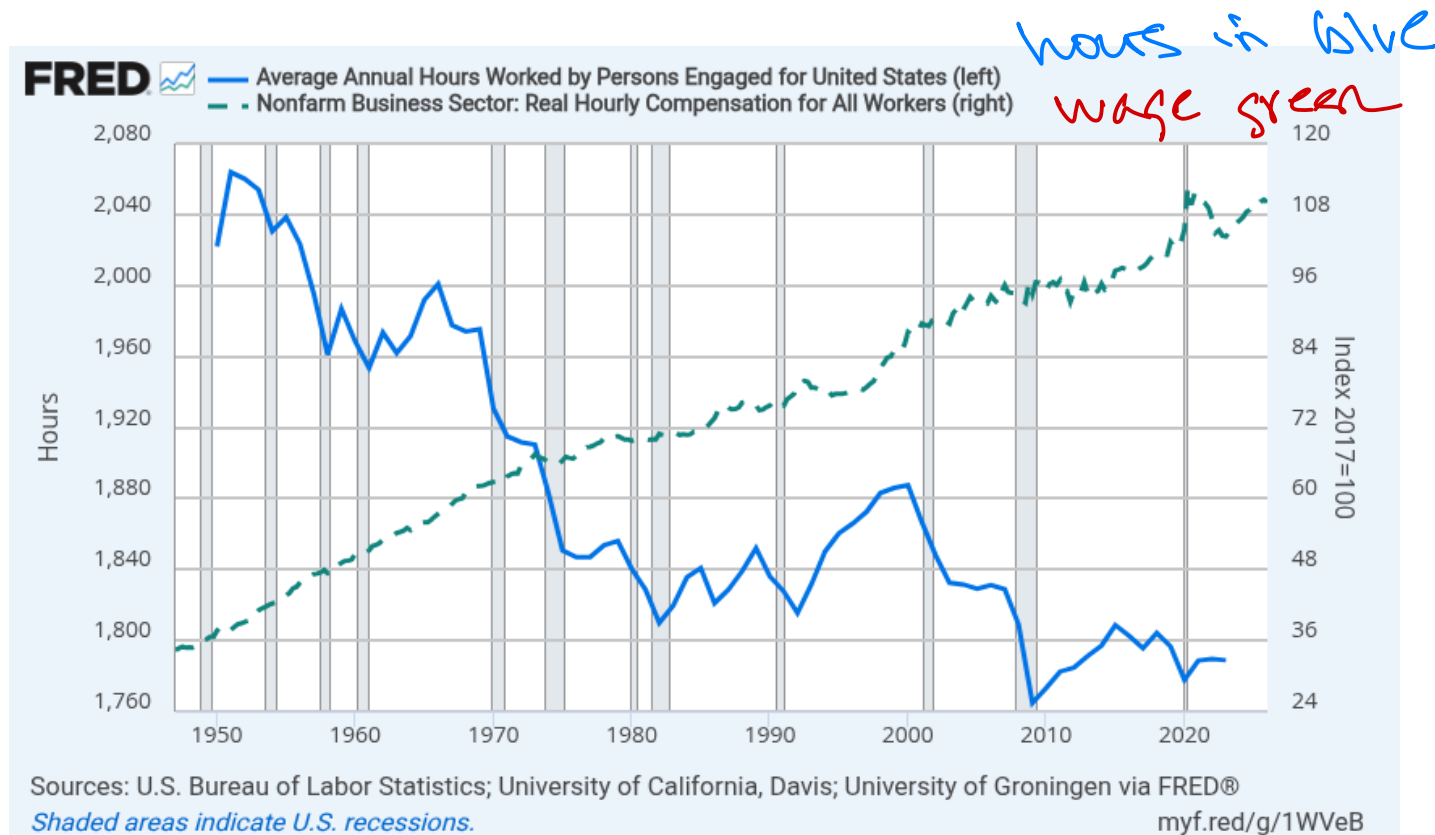


Movements at the extensive margin seem driven by something other than real returns to employment



wages $\uparrow$ $\Rightarrow$ want more leisure, less hours
at work (income effect)

**Income effects appear to dominate substitution effects
at the intensive margin over long periods of time**



* People increasingly have multiple part-time jobs so hours per job has fallen but not

But average annual hours is not the same as hours per worker (Mcgratten and Rogerson). Hours per worker has only fallen 5%:

per worker.

Table 1

A Look Behind an Aggregate Fact

In the United States, 1950–2000

Year	Average Weekly Hours Worked		Employment-to-Population Ratio, in %
	Per Person	Per Worker	
1950	22.34	42.40	52.69
1960	21.55	40.24	53.55
1970	21.15	38.83	54.47
1980	22.07	39.01	56.59
1990	23.86	39.74	60.04
2000	23.94	40.46	59.17
% Change 1950–2000	7.18	-4.56	12.30

Source: U.S. Department of Commerce, Bureau of the Census

Table 1 also provides a decomposition of hours worked per person into weekly hours worked per worker (what economists call the *intensive margin*) and the percentage of the population aged 15 and over employed (the *extensive margin*).³ Note that average weekly hours worked per worker are not necessarily the same as the average length of the workweek across jobs. That's because many individuals hold multiple jobs, and average weekly hours aggregate these hours across jobs. For example, if a worker has two jobs, each of which has a workweek of 20 hours, then that worker reports weekly hours of 40, and not the workweek of the individual jobs.

Smaller decline than in the graph we just saw.



Summary: hours pretty flat

Table 2

Distributions of Hours Worked by Sex

Average Weekly Hours Worked per Person and per Worker
and the Employment-to-Population Ratio in the United States, 1950–2000

Year	Total Population	Hours Worked per Person	
		Sex	
		Males	Females
1950	22.34	34.18	10.87
1960	21.55	31.93	11.84
1970	21.15	29.75	13.38
1980	22.07	28.70	16.02
1990	23.86	29.11	19.03
2000	23.94	28.34	19.78
% Change 1950–2000	7.18	-17.09	81.98

Year	Total Population	Hours Worked per Worker	
		Sex	
		Males	Females
1950	42.40	44.00	38.19
1960	40.24	42.45	35.67
1970	38.83	41.33	34.70
1980	39.01	41.84	35.10
1990	39.74	42.59	36.30
2000	40.46	43.27	37.15
% Change 1950–2000	-4.56	-1.65	-2.73

Year	Total Population	Employment-to-Population Ratio, in %	
		Sex	
		Males	Females
1950	52.69	77.69	28.46
1960	53.55	75.22	33.20
1970	54.47	71.98	38.55
1980	56.59	68.60	45.64
1990	60.04	68.35	52.42
2000	59.17	65.49	53.25
% Change 1950–2000	12.30	-15.70	87.10

Source: U.S. Department of Commerce, Bureau of the Census

Most of the small remaining aggregate dynamics at the intensive margin are compositional effects, as women, who work on average fewer hours, make up more of the labor force.

↓ 5% ↓
Balances out to -2.73% small
women
of make up more
and 1/2 force
more likely
part-time



extensive margin

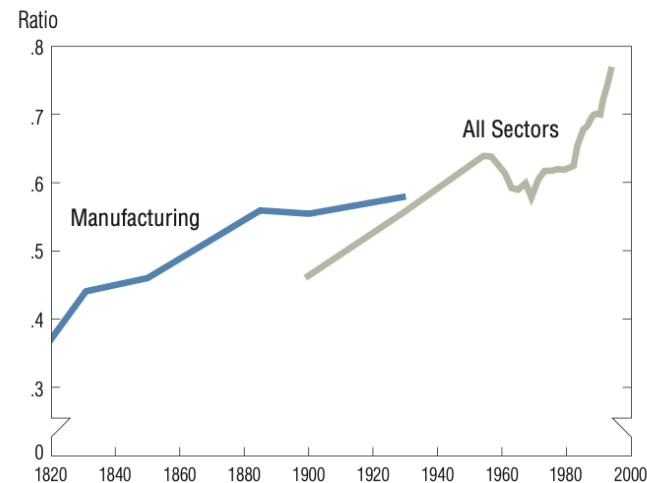
Increased female labor force participation may reflect strong female-specific after-tax real wage increases (substitution effect)

One factor relevant to the reallocation of hours from males to females is the narrowing of the gap between female and male wages. In Chart 4, we plot the ratio of female to male before-tax earnings for full-time workers in manufacturing between 1820 and 1930 and in all sectors between 1900 and 1992. The source of these data is Goldin 1994. As the chart shows, the gap has narrowed considerably. This could be the consequence of less gender discrimination or increased relative productivity of females as the importance of physical strength to work has waned. If we take into account changes in U.S. tax policy that led to lower effective marginal tax rates on wives, then we find a further narrowing of the gender wage gap. Jones, Manuelli, and McGrattan (2003) argue that observed changes in the gender wage gap can explain simultaneously the significant increases in average hours worked by married women and the relative constancy in the hours worked by single women, single men, and married men.

Chart 4

A Narrowing of the Gender Wage Gap

The Ratio of Female to Male Before-Tax Earnings for Full-Time Workers in Manufacturing Between 1820 and 1930 and in All Sectors Between 1900 and 1992, in the United States



Source: Goldin 1994

(wage ↑ for women
↓ for men)





Summary so far:

A breakdown of the aggregate intensive margin data could be interpreted as either

- ⇒ Little-to-no sensitive of hours worked to changes in total real compensation (income and substitution effects cancel over the long run), or
- ⇒ Substitution effects being strong enough to dominate income effects for women, reflecting greater gender equality in the workforce as well reduction of the “marriage tax” in the US tax code.

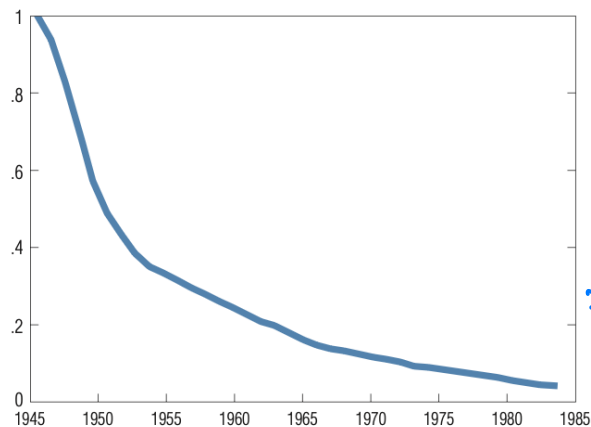
(extensive margin)

Increased female labor force participation may reflect a higher relative weight on “C” in female utility

Chart 5

Innovating in Home Technology

The Price Index for Eight Home Appliances
Relative to the GDP Deflator, 1947–84



Source: Gordon 1990

Table 11

Changes in Family Structure

Percentage of the Population in Each Marital Status Category
In the United States, 1950–2000

Year	Married With Spouse		Single	Widowed	Divorced
	Present	Absent			
1950	64.00	4.43	21.08	8.25	2.24
1960	65.59	3.71	20.12	8.03	2.56
1970	61.42	3.72	23.24	8.19	3.43
1980	56.78	3.43	25.95	7.64	6.19
1990	53.60	4.10	26.43	7.48	8.39
2000	51.26	5.27	27.09	6.63	9.75
% Change 1950–2000	-19.91	18.96	28.51	-19.64	335.27

Source: U.S. Department of Commerce, Bureau of the Census

separated, single, widowed, or divorced. If marital status is a significant determinant of hours of work, these shifts in family structure should imply significant changes in work hours.

more women without a
second male income



Better retirement benefits $\Rightarrow$ $\uparrow$ $\uparrow$
 income effect
 after age 65

Table 4 Males

Year	Hours Worked per Person by Age (in Years)						
	15-24	25-34	35-44	45-54	55-64	65-74	75-99
1950	24.32	39.84	41.82	40.17	35.35	20.74	7.39
1960	20.39	40.01	41.01	39.40	33.65	13.32	5.15
1970	18.46	39.55	40.86	39.19	32.78	10.77	3.65
1980	20.00	37.93	40.18	38.04	28.94	7.65	2.62
1990	19.01	37.96	39.83	38.60	27.06	7.30	2.23
2000	17.74	36.39	37.93	37.19	27.20	7.69	2.70
% Change 1950-2000	-27.05	-8.64	-9.30	-7.41	-23.07	-62.92	-63.50

Table 5 Females

Year	Hours Worked per Person by Age (in Years)						
	15-24	25-34	35-44	45-54	55-64	65-74	75-99
1950	12.36	11.76	13.07	12.47	8.77	3.82	1.00
1960	10.77	11.90	14.71	16.49	12.31	4.23	1.37
1970	12.39	15.13	17.10	18.46	14.78	3.95	1.27
1980	15.20	22.29	22.13	20.41	14.16	3.17	0.81
1990	15.63	26.41	27.62	26.06	15.67	3.54	0.70
2000	14.84	26.76	27.33	28.00	18.21	4.09	0.88
% Change 1950-2000	20.05	127.58	109.15	124.53	107.52	7.11	-11.83

At the extensive margin, there has been a shift in the age of people at work, with more young women working and fewer older men working

Decrease



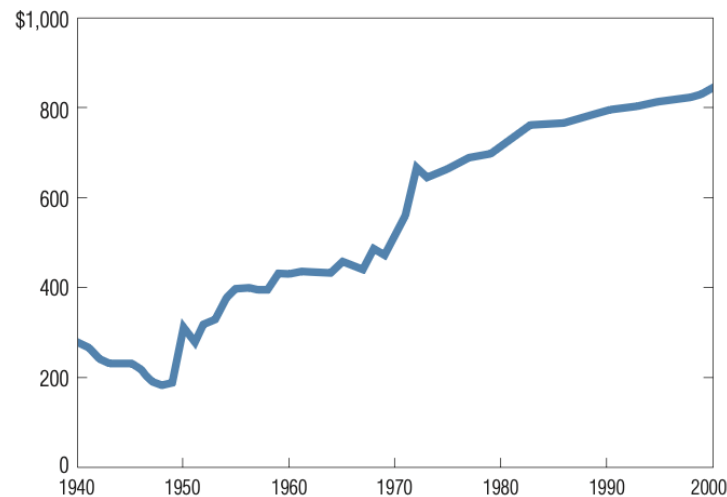
Lower older-age participation, especially among men, may reflect higher retirement benefits (income effect)



Chart 6

Increasing Retirement Benefits

Average Monthly Social Security Benefits to U.S.
Retired Workers in Constant 2000 Dollars, 1940–2000



Source: U.S. Social Security Administration

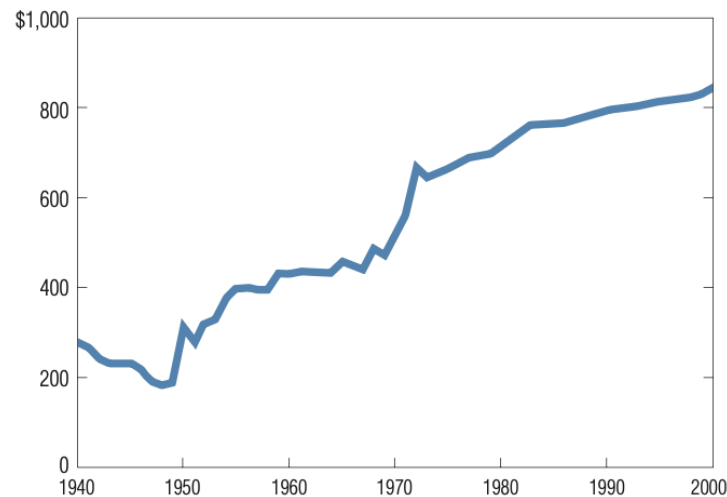
↑↑↑
positive
income
effect
C ↑ L ↑
N ↓

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